Chapter 03 | Functions of a real variable

Worked solutions

Original, unofficial exercises. Source pages guide conceptual coverage only. Always check the domain assumptions.

Exam 1 | 80 minutes | 48 marks

24 problems (2 marks each). Try without notes or a calculator.

L37-P041 | Find the domain of f(x) = (x^2 + 1)/((x - 2)(x + 1)).

The denominator is zero for x = 2 or x = -1. Domain: all real numbers except those values.

L37-P048 | Find the domain of f(x) = (x^2 + 4)/((x - 4)(x + 4)).

The denominator is zero for x = 4 or x = -4. Domain: all real numbers except those values.

L37-P055 | Find the domain of f(x) = (x^2 + 3)/((x - 6)(x + 3)).

The denominator is zero for x = 6 or x = -3. Domain: all real numbers except those values.

L37-P062 | f(x) = 3x - 3. Solve f(x) = -6 and state the corresponding graph point.

3x - 3 = -6, so 3x = -3 and x = -1. Point: (-1, -6).

L38-P041 | Using the difference quotient limit, find f’(-2) for f(x) = 1x^2 - 3x.

[f(p+h)-f(p)]/h = 1(2p+h) - 3; as h→0: 1×2×(-2) - 3 = -7.

L38-P048 | Using the difference quotient limit, find f’(0) for f(x) = 4x^2 - 3x.

[f(p+h)-f(p)]/h = 4(2p+h) - 3; as h→0: 4×2×(0) - 3 = -3.

L38-P055 | Using the difference quotient limit, find f’(2) for f(x) = 3x^2 - 3x.

[f(p+h)-f(p)]/h = 3(2p+h) - 3; as h→0: 3×2×(2) - 3 = 9.

L38-P062 | f(x) = 2x^2 - 2x. Find the tangent-line equation at x = -1.

f(-1) = 4; f’(-1) = -6. Tangent: y = 4 + (-6)(x - (-1)).

L39-P041 | Differentiate f(x) = (1x - 3)(1x - 3).

Set u = 1x - 3, v = 1x - 3. f’ = 1(1x - 3) + (1x - 3)1 = 2*x - 6.

L39-P048 | Differentiate f(x) = (3x - 3)(4x - 3).

Set u = 3x - 3, v = 4x - 3. f’ = 3(4x - 3) + (3x - 3)4 = 24*x - 21.

L39-P055 | Differentiate f(x) = (5x - 3)(3x - 3).

Set u = 5x - 3, v = 3x - 3. f’ = 5(3x - 3) + (5x - 3)3 = 30*x - 24.

L39-P062 | Differentiate f(x) = (x^2 + 3)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 3)]/(x - 2)^2 = (x^2 - 4x - 3)/(x - 2)^2.

L40-P041 | Differentiate f(x) = 2e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 2e^(1x + 1).

L40-P048 | Differentiate f(x) = 4e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 16e^(4x + 4).

L40-P055 | Differentiate f(x) = 6e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 18e^(3x + 3).

L40-P062 | Solve 3e^x = 3e^2 for real x.

Divide by nonzero 3: e^x=e^2; exponential is one-to-one, so x=2.

L41-P041 | For f(x)=ln(x), compute f’(1) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(1)=1/1.

L41-P048 | For f(x)=ln(x), compute f’(4) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(4)=1/4.

L41-P055 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P062 | Differentiate f(x)=ln(3x + 2) and find f’(2); state the domain.

Require 3x + 2 > 0, so x > -2/3. Chain rule: f’(x)=3/(3x + 2); f’(2)=3/8.

L42-P041 | Find log_2(2) using change of base.

log_2(2)=ln(2)/ln(2)=1ln(2)/ln(2)=1.

L42-P048 | Find log_5(25) using change of base.

log_5(25)=ln(25)/ln(5)=2ln(5)/ln(5)=2.

L42-P055 | Find log_4(64) using change of base.

log_4(64)=ln(64)/ln(4)=3ln(4)/ln(4)=3.

L42-P062 | Differentiate f(x)=log_3(3x + 3) and find f’(2); state the domain.

Domain x > -3/3. Since log_3(u)=ln(u)/ln(3), f’(x)=3/[(3x + 3) ln(3)]; at 2: 1/3/ln(3).

Exam 2 | 80 minutes | 48 marks

24 problems (2 marks each). Try without notes or a calculator.

L37-P047 | Find the domain of f(x) = (x^2 + 3)/((x - 3)(x + 3)).

The denominator is zero for x = 3 or x = -3. Domain: all real numbers except those values.

L37-P054 | Find the domain of f(x) = (x^2 + 2)/((x - 5)(x + 2)).

The denominator is zero for x = 5 or x = -2. Domain: all real numbers except those values.

L37-P061 | f(x) = 2x - 4. Solve f(x) = -10 and state the corresponding graph point.

2x - 4 = -10, so 2x = -6 and x = -3. Point: (-3, -10).

L37-P068 | f(x) = 4x + 3. Solve f(x) = -9 and state the corresponding graph point.

4x + 3 = -9, so 4x = -12 and x = -3. Point: (-3, -9).

L38-P047 | Using the difference quotient limit, find f’(-1) for f(x) = 3x^2 + 3x.

[f(p+h)-f(p)]/h = 3(2p+h) + 3; as h→0: 3×2×(-1) + 3 = -3.

L38-P054 | Using the difference quotient limit, find f’(1) for f(x) = 2x^2 + 3x.

[f(p+h)-f(p)]/h = 2(2p+h) + 3; as h→0: 2×2×(1) + 3 = 7.

L38-P061 | f(x) = 1x^2 - 3x. Find the tangent-line equation at x = -2.

f(-2) = 10; f’(-2) = -7. Tangent: y = 10 + (-7)(x - (-2)).

L38-P068 | f(x) = 4x^2 - 3x. Find the tangent-line equation at x = 0.

f(0) = 0; f’(0) = -3. Tangent: y = 0 + (-3)(x - (0)).

L39-P047 | Differentiate f(x) = (2x + 3)(3x + 1).

Set u = 2x + 3, v = 3x + 1. f’ = 2(3x + 1) + (2x + 3)3 = 12*x + 11.

L39-P054 | Differentiate f(x) = (4x + 3)(2x + 1).

Set u = 4x + 3, v = 2x + 1. f’ = 4(2x + 1) + (4x + 3)2 = 16*x + 10.

L39-P061 | Differentiate f(x) = (x^2 + 2)/(x - 1), stating its domain.

Domain x ≠ 1. Quotient rule: f’ = [2x(x - 1) - (x^2 + 2)]/(x - 1)^2 = (x^2 - 2x - 2)/(x - 1)^2.

L39-P068 | Differentiate f(x) = (x^2 + 3)/(x - 3), stating its domain.

Domain x ≠ 3. Quotient rule: f’ = [2x(x - 3) - (x^2 + 3)]/(x - 3)^2 = (x^2 - 6x - 3)/(x - 3)^2.

L40-P047 | Differentiate f(x) = 3e^(3x + 3).

By the chain rule D(e^u)=u’e^u, where u’=3. Hence f’(x) = 9e^(3x + 3).

L40-P054 | Differentiate f(x) = 5e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 10e^(2x + 2).

L40-P061 | Solve 2e^x = 2e^1 for real x.

Divide by nonzero 2: e^x=e^1; exponential is one-to-one, so x=1.

L40-P068 | Solve 4e^x = 4e^4 for real x.

Divide by nonzero 4: e^x=e^4; exponential is one-to-one, so x=4.

L41-P047 | For f(x)=ln(x), compute f’(3) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(3)=1/3.

L41-P054 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P061 | Differentiate f(x)=ln(2x + 1) and find f’(1); state the domain.

Require 2x + 1 > 0, so x > -1/2. Chain rule: f’(x)=2/(2x + 1); f’(1)=2/3.

L41-P068 | Differentiate f(x)=ln(4x + 4) and find f’(4); state the domain.

Require 4x + 4 > 0, so x > -4/4. Chain rule: f’(x)=4/(4x + 4); f’(4)=1/5.

L42-P047 | Find log_4(4) using change of base.

log_4(4)=ln(4)/ln(4)=1ln(4)/ln(4)=1.

L42-P054 | Find log_3(9) using change of base.

log_3(9)=ln(9)/ln(3)=2ln(3)/ln(3)=2.

L42-P061 | Differentiate f(x)=log_2(2x + 2) and find f’(1); state the domain.

Domain x > -2/2. Since log_2(u)=ln(u)/ln(2), f’(x)=2/[(2x + 2) ln(2)]; at 1: 1/2/ln(2).

L42-P068 | Differentiate f(x)=log_5(5x + 3) and find f’(4); state the domain.

Domain x > -3/5. Since log_5(u)=ln(u)/ln(5), f’(x)=5/[(5x + 3) ln(5)]; at 4: 5/23/ln(5).

Exam 3 | 80 minutes | 48 marks

24 problems (2 marks each). Try without notes or a calculator.

L37-P046 | Find the domain of f(x) = (x^2 + 2)/((x - 2)(x + 2)).

The denominator is zero for x = 2 or x = -2. Domain: all real numbers except those values.

L37-P053 | Find the domain of f(x) = (x^2 + 1)/((x - 4)(x + 1)).

The denominator is zero for x = 4 or x = -1. Domain: all real numbers except those values.

L37-P060 | Find the domain of f(x) = (x^2 + 4)/((x - 6)(x + 4)).

The denominator is zero for x = 6 or x = -4. Domain: all real numbers except those values.

L37-P067 | f(x) = 3x + 2. Solve f(x) = 8 and state the corresponding graph point.

3x + 2 = 8, so 3x = 6 and x = 2. Point: (2, 8).

L38-P046 | Using the difference quotient limit, find f’(-2) for f(x) = 2x^2 + 2x.

[f(p+h)-f(p)]/h = 2(2p+h) + 2; as h→0: 2×2×(-2) + 2 = -6.

L38-P053 | Using the difference quotient limit, find f’(0) for f(x) = 1x^2 + 2x.

[f(p+h)-f(p)]/h = 1(2p+h) + 2; as h→0: 1×2×(0) + 2 = 2.

L38-P060 | Using the difference quotient limit, find f’(2) for f(x) = 4x^2 + 2x.

[f(p+h)-f(p)]/h = 4(2p+h) + 2; as h→0: 4×2×(2) + 2 = 18.

L38-P067 | f(x) = 3x^2 + 3x. Find the tangent-line equation at x = -1.

f(-1) = 0; f’(-1) = -3. Tangent: y = 0 + (-3)(x - (-1)).

L39-P046 | Differentiate f(x) = (1x + 2)(2x - 2).

Set u = 1x + 2, v = 2x - 2. f’ = 1(2x - 2) + (1x + 2)2 = 4*x + 2.

L39-P053 | Differentiate f(x) = (3x + 2)(1x - 2).

Set u = 3x + 2, v = 1x - 2. f’ = 3(1x - 2) + (3x + 2)1 = 6*x - 4.

L39-P060 | Differentiate f(x) = (5x + 2)(4x - 2).

Set u = 5x + 2, v = 4x - 2. f’ = 5(4x - 2) + (5x + 2)4 = 40*x - 2.

L39-P067 | Differentiate f(x) = (x^2 + 2)/(x - 2), stating its domain.

Domain x ≠ 2. Quotient rule: f’ = [2x(x - 2) - (x^2 + 2)]/(x - 2)^2 = (x^2 - 4x - 2)/(x - 2)^2.

L40-P046 | Differentiate f(x) = 2e^(2x + 2).

By the chain rule D(e^u)=u’e^u, where u’=2. Hence f’(x) = 4e^(2x + 2).

L40-P053 | Differentiate f(x) = 4e^(1x + 1).

By the chain rule D(e^u)=u’e^u, where u’=1. Hence f’(x) = 4e^(1x + 1).

L40-P060 | Differentiate f(x) = 6e^(4x + 4).

By the chain rule D(e^u)=u’e^u, where u’=4. Hence f’(x) = 24e^(4x + 4).

L40-P067 | Solve 3e^x = 3e^3 for real x.

Divide by nonzero 3: e^x=e^3; exponential is one-to-one, so x=3.

L41-P046 | For f(x)=ln(x), compute f’(2) and state the domain.

The domain is x>0; f’(x)=1/x, thus f’(2)=1/2.

L41-P053 | For f(x)=ln(x), compute f′(1) and state the domain.

The domain is x>0; f′(x)=1/x, thus f′(1)=1/1.

L41-P060 | For f(x)=ln(x), compute f′(4) and state the domain.

The domain is x>0; f′(x)=1/x, thus f′(4)=1/4.

L41-P067 | Differentiate f(x)=ln(3x + 3) and find f′(3); state the domain.

Require 3x + 3 > 0, so x > -3/3. Chain rule: f′(x)=3/(3x + 3); f′(3)=1/4.

L42-P046 | Find log_3(729) using change of base.

log_3(729)=ln(729)/ln(3)=6ln(3)/ln(3)=6.

L42-P053 | Find log_2(2) using change of base.

log_2(2)=ln(2)/ln(2)=1ln(2)/ln(2)=1.

L42-P060 | Find log_5(25) using change of base.

log_5(25)=ln(25)/ln(5)=2ln(5)/ln(5)=2.

L42-P067 | Differentiate f(x)=log_4(4x + 2) and find f′(3); state the domain.

Domain x > -2/4. Since log_4(u)=ln(u)/ln(4), f′(x)=4/[(4x + 2) ln(4)]; at 3: 2/7/ln(4).